

Cost-effectiveness of gargling for the prevention of upper respiratory tract infections

Background In Japan, gargling is a generally accepted way of preventing upper respiratory tract infection (URTI). The effectiveness of gargling for preventing URTI has been shown in a randomized controlled trial that compared incidences of URTI between gargling and control groups. From the perspective of the third-party payer, gargling is dominant due to the fact that the costs of gargling are borne by the participant. However, the cost-effectiveness of gargling from a societal perspective should be considered. In this study, economic evaluation alongside a randomized controlled trial was performed to evaluate the cost-effectiveness of gargling for preventing URTI from a societal perspective. **Methods** Among participants in the gargling trial, 122 water-gargling and 130 control subjects were involved in the economic analysis. Sixty-day cumulative follow-up costs and effectiveness measured by quality-adjusted life days (QALD) were compared between groups on an intention-to-treat basis. Incremental cost-effectiveness ratio (ICER) was converted to dollars per quality-adjusted life years (QALY). The 95% confidence interval (95%CI) and probability of gargling being cost-effective were estimated by bootstrapping. **Results** After 60 days, QALD was increased by 0.43 and costs were \$37.1 higher in the gargling group than in the control group. ICER of the gargling group was \$31,800/QALY (95%CI, \$1,900–\$248,100). Although this resembles many acceptable forms of medical intervention, including URTI preventive measures such as influenza vaccination, the broad confidence interval indicates uncertainty surrounding our results. In addition, one-way sensitivity analysis also indicated that careful evaluation is required for the cost of gargling and the utility of moderate URTI. The major limitation of this study was that this trial was conducted in winter, at a time when URTI is prevalent. Care must be taken when applying the results to a season when URTI is not prevalent, since the ICER will increase due to decreases in incidence. **Conclusion** This study suggests gargling as a cost-effective preventive strategy for URTI that is acceptable from perspectives of both the third-party payer and society.